CRIMPSTON SOFTWARE ENGINEERING

RTR Wholesale Management

Professional Project Overview, Technology Stack, Features & Technical Architecture

Official Client Handover Documentation

Client Organization:	**RTR Mobiles**
Engineering Partner:	**Crimpston**
Project Classification:	**Wholesale Business Management Platform**
Release Version:	**1.0.0 (Production Go-Live Release)**
Publication Date:	**November 7, 2026**

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1. Executive Summary & Core Objectives

RTR Wholesale Management is an enterprise-grade digital business management platform specifically engineered for wholesale mobile accessory distribution across multi-store retail partner networks. Developed by Crimpston for RTR Mobiles, the system modernizes and unifies day-to-day wholesale operations into a centralized, real-time platform.

By replacing manual paper ledgers and fragmented spreadsheets with a modern web architecture, RTR Wholesale Management empowers business owners and authorized retail staff with complete visibility into inventory dispatches, shop-by-shop sales transactions, delivery statuses, payment collections, and net profitability.

Key Business Objectives Delivered:

- Centralizes partner shop accounts, delivery logs, and outstanding balances in a single reliable system.
- Eliminates calculation discrepancies with automated profit formulas: *Profit = Customer Price - Wholesale Amount*.
- Implements granular access control ensuring retail employees cannot view company-wide margins or other shops' data.
- Guarantees high-speed access across mobile and desktop devices with optimized cloud hosting.

2. Verified Technology Stack

Technology	Component Layer	Purpose & Business Value
React (v18)	Frontend Interface	Component-driven user interface delivering instant page rendering, real-time modal updates, and responsive mobile layout.
TypeScript	Language / Type Safety	Strict end-to-end type safety eliminating runtime bugs and ensuring accurate mathematical calculations across all transactions.
Vite	Build & Tooling	High-performance frontend bundler providing rapid page loads, code splitting, and automated production asset optimization.
Node.js & Express	Backend API Service	Robust, asynchronous API gateway enforcing rate limiting, request validation, transaction timeouts, and catalog caching.
Supabase & Postgres	Database & Storage	Enterprise relational database providing ACID transaction compliance, foreign key integrity, and Row-Level Security (RLS).
Supabase Auth	Authentication Engine	Cryptographically secure token management, session persistence, and self-service password recovery workflows.
Vercel	Frontend Hosting	Global Edge Content Delivery Network (CDN) ensuring 99.99% availability, automated SSL encryption, and low latency.

Render	Backend Hosting	Containerized cloud infrastructure hosting the API server with automated health probes and continuous zero-downtime deploys.

The application follows a modern decoupled architecture built upon industry-standard, production-proven technologies:

3. Application Features & Business Capabilities

1. Executive Dashboard

Provides the business owner with an immediate high-level summary of total wholesale revenue, collected payments, outstanding market credit, and dynamic gross profit. Offers one-click action buttons to record new deliveries, register shops, or update catalog items.

2. Shop & Retail Partner Management

Maintains an active directory of all retail shop partners with contact details, delivery records, and live account balances. Includes safe-deletion protection preventing accidental deletion of shops with historical delivery records.

3. Master Product Catalog Management

Empowers administrators to maintain product lines, SKU numbers, and baseline wholesale pricing. Allows active/inactive status toggling so discontinued items remain archived without corrupting past invoices.

4. Stock Delivery & Sales Recording

Fast, single-screen transaction drawer accessible via `Alt+N`. Captures Date, Retail Shop, Product, Quantity (Pieces), Wholesale Amount (₹), Delivery Status, and Customer Price (CP). Features real-time profit reflection and automated selling-below-cost warnings.

5. Payment Management & Outstanding Balances

Tracks full, partial, and installment payments against partner shop accounts. Automatically updates closing balances across statements and preserves complete chronological audit logs of all cash and digital settlements.

6. Statements, Financial Reporting & Export

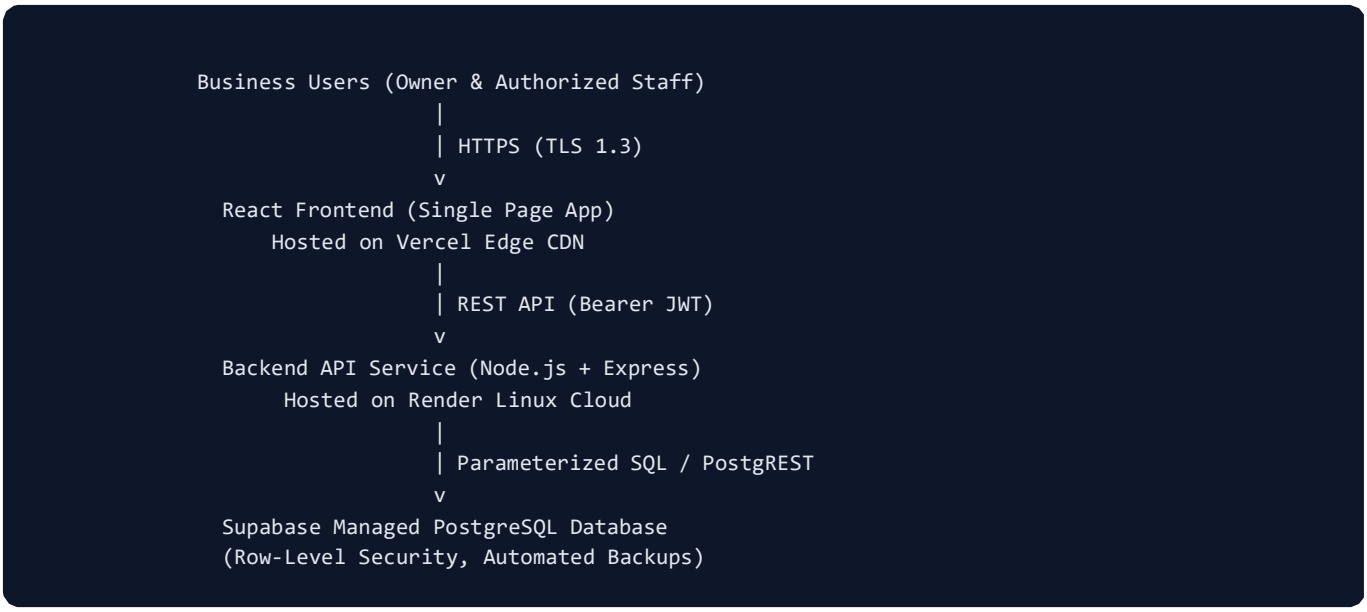
Generates detailed Monthly Shop Statements and deep-dive Individual Shop Statements. Features one-click formatted export to Excel (.xlsx) and CSV files, alongside print-ready invoice layouts for physical billing.

7. Role-Based Employee Access

Enforces strict role separation: Business Owners access global analytics, pricing, and all shop records, while retail floor Employees are strictly sandboxed to view only their assigned shop location in read-only mode.

4.System Architecture

The system is architected across four distinct, decoupled tiers to guarantee security, modularity, and operational independence:



5.Scalability, Traffic Control & Performance

Specialized scalability and reliability engineering has been incorporated directly into the API and database tiers:

- **Tiered API Rate Limiting:** Enforces sliding-window traffic limits protecting against abuse (30 req/15 min for authentication, 60 req/15 min for financial mutations, 300 req/15 min for standard read endpoints).
- **Request Timeout & Disconnect Protection:** Enforces a 25-second server ceiling. If a mobile user loses connection in a warehouse, background queries are instantly aborted via connection listener hooks, conserving database compute.
- **Financial Idempotency Protection:** Financial mutations support `Idempotency-Key` headers, preventing duplicate charge or ledger entries if a user accidentally double-clicks submit.
- **High-Speed Catalog In-Memory Caching:** Serves shop and product directories with a 60-second Time-To-Live (TTL) and instant cache invalidation upon any catalog modification.
- **Database Indexing:** Migration `0012` introduces composite B-tree indexes across high-volume transaction tables, ensuring near-instantaneous monthly statement calculations.

> **Performance Testing Note:** Synthetic load testing was executed against a local API server harness across 5, 25, 50, and 100 concurrent virtual users to verify throttling and error handling. Actual live cloud capacity will reflect the provisioned Render and Supabase cloud tiers.

6.Security, Authentication & Access Control

The platform adheres strictly to modern web application security principles:

- **Row-Level Security (RLS):** PostgreSQL engine-level policies strictly isolate business records and enforce shop-level access controls directly at the database tier.
- **Zero Secret Leakage:** Client applications access APIs exclusively with public identifiers and temporary JWT sessions. Database service-role secrets and private keys are never exposed in client code.
- **Automated Log Sanitization:** All server terminal outputs and browser console messages are processed through `safeLogger.ts`, automatically masking passwords, tokens, and customer UUIDs.

7.Email & Password Recovery System

The application utilizes the **Supabase Default Mailer service** to dispatch account recovery instructions. When a user requests a password reset from `/forgot-password`, a time-limited secure link directs them to the canonical reset form (`https://rtr-management.vercel.app/reset-password`) to establish a new password.

8.Hosting & Production Deployment

Service Component	Hosting Provider	Production Endpoint
Frontend Web Application	Vercel Global CDN	`https://rtr-management.vercel.app`
Backend REST API	Render Cloud Platform	`https://rtr-management.onrender.com`
Database & Authentication	Supabase Managed Cloud	Authoritative PostgreSQL Instance

9.Future Expansion Roadmap

The system's modular architecture enables straightforward future additions, including distributed Redis caching, dedicated enterprise SMTP (SendGrid/AWS SES), automated WhatsApp dispatch receipts, barcode scanning integration, and predictive stock re-ordering.

Prepared by Crimpston Engineering Team | Official Client Release Document.